

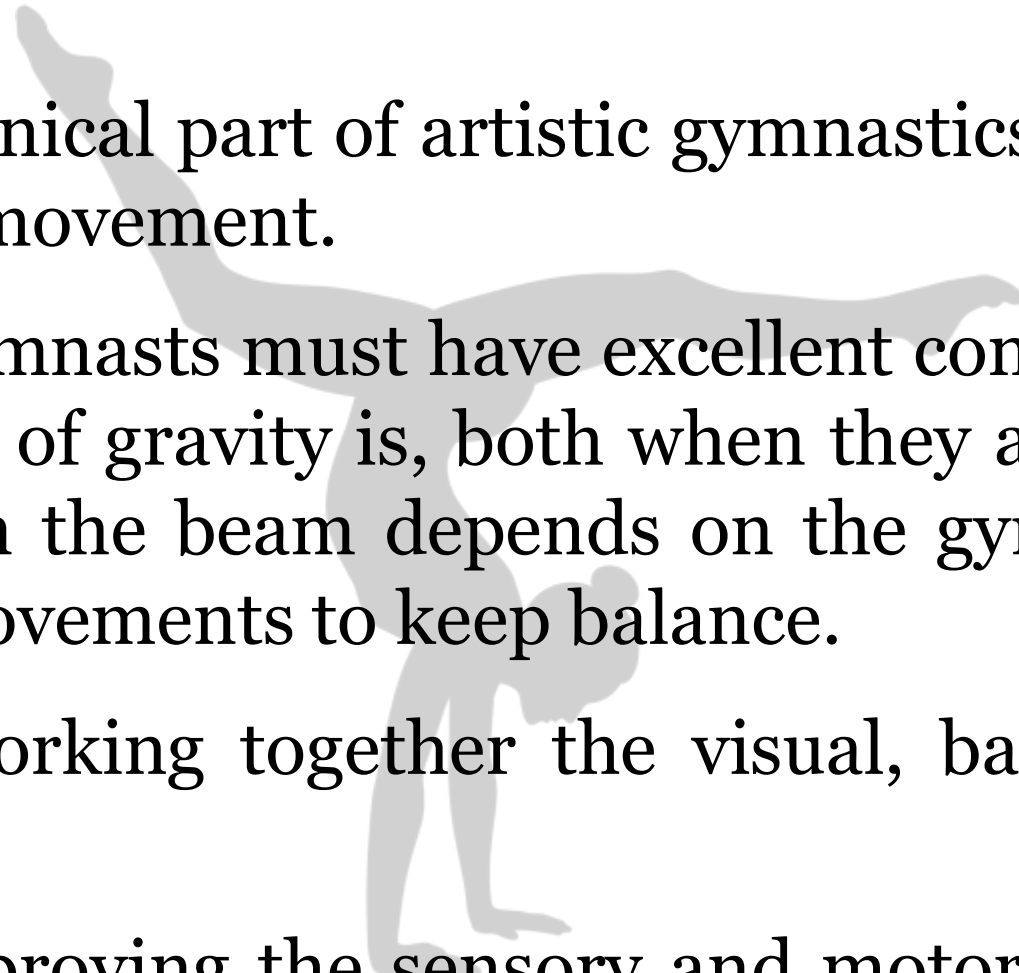


EFFECT OF SENSORIMOTOR FOOT TRAINING ON BALANCE BEAM STABILITY IN YOUNG GYMNASTS

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INTRODUCTION

- ▶ The balance beam is a very technical part of artistic gymnastics that needs great control over body position and movement.
- ▶ The beam is very narrow, so gymnasts must have excellent control of their body position and where their center of gravity is, both when they are still and when they are moving. Doing well on the beam depends on the gymnast's ability to match their senses with their movements to keep balance.
- ▶ Keeping balance is done by working together the visual, balance, and touch systems in the body.
- ▶ Recent studies suggest that improving the sensory and motor skills of the feet may help improve balance and stability.



METHODOLOGY

- ▶ **Study Design** : This investigation utilized a cross-sectional observational approach to evaluate postural stability and foot sensorimotor capabilities in young female gymnasts during balance beam tasks.
- ▶ **Participants** : The study included 30 young female gymnasts aged 8 to 14 who had participated in gymnastics for at least two years. All participants were actively training on the balance beam and had no lower limb injuries for the preceding six months. Informed consent was secured from the parents or guardians.

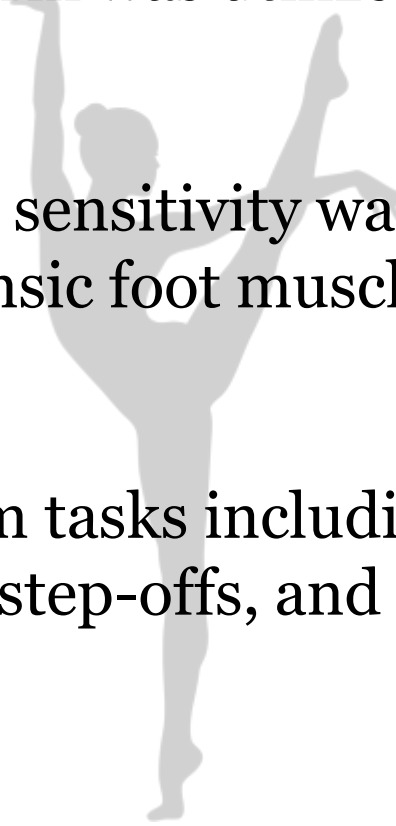
Assessment Tools : Balance Beam Test: Standardized balance beam tasks, including single-leg static stances, walking, and turning movements, were performed. Instances of balance errors, such as wobbles, step-offs, and falls, were documented.

I. Postural Stability Assessment: A force platform was utilized to measure postural sway parameters during single-leg stance tasks.

II. Foot Sensorimotor Function Tests: - Plantar sensitivity was tested using monofilaments. - Proprioception was assessed. - Evaluation of intrinsic foot muscle strength was conducted.

III. Balance Beam Test :

Participants performed standardized balance beam tasks including static single-leg stance, walking, and turn movements. Balance errors such as wobbles, step-offs, and falls were recorded.





Procedure : Participants conducted the balance tasks without footwear to ensure accurate sensory feedback from their feet. Each task was executed three times, and the mean value was calculated for analysis, with sufficient rest intervals between trials.

Statistical Analysis : Descriptive statistics were calculated for each variable, and correlation analyses were conducted to explore the relationships between foot sensorimotor functions and postural stability measures. Statistical significance was established at $p < 0.05$.

RESULT AND DISCUSSION

TABLE 1 :PARTICIPANT CHARACTERISTICS

VARIABLE	MEAN
AGE	20 - 22
TRAINING EXPERENC E (YEARS)	1 - 3.5
WEEKLY TRAINING HOURS	3.2- 12.5

TABLE 2 : POSTURAL STABILITY MEASURES

PARAMET ERS	MEAN
Anterior-posterior sway(cm)	0.6 - 2.8
Mediolate ral sway(cm)	0.5 - 2.4
Total sway area(cm)	1.1 - 5.9

TABLE 3: FOOT SENSORIMOTOR FUNCTION

VARIABLE	MEAN
Plantar sensitivity score	20 - 22
Propriocep tive error	0.9 - 3.1
Intrinsic - foot muscle strength	1.0 - 7.8

TABLE 4 : BALANCE ERRORS
DURING BEAM PERFORMANCE

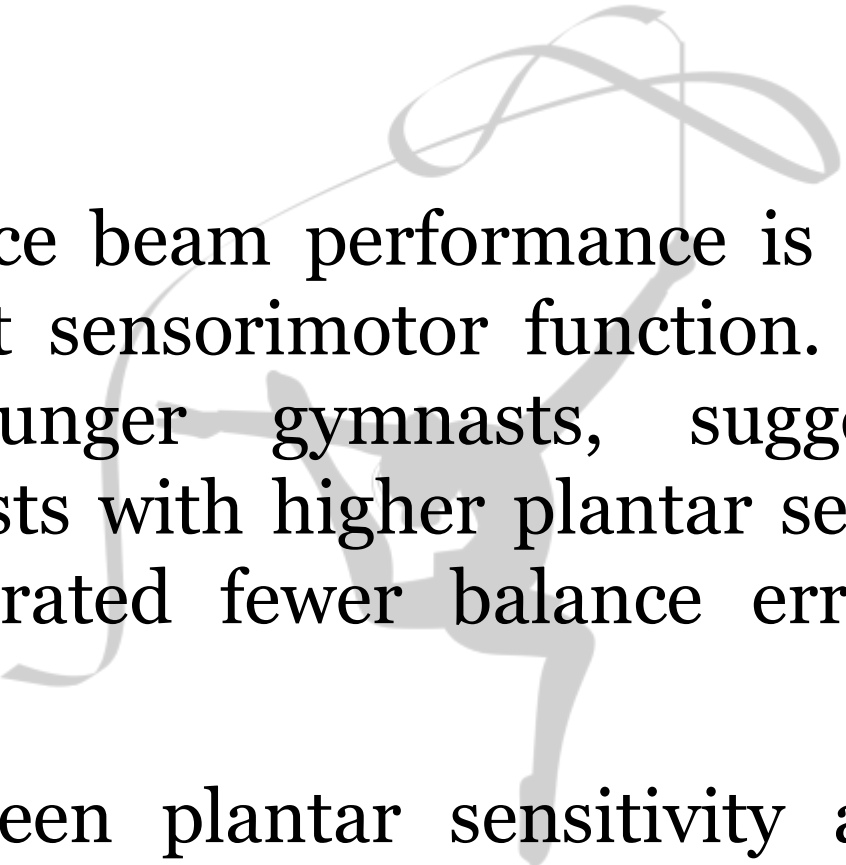
ERROR TYPE	FREQUEN CY
WOBBLES	0.3 - 2.3
STEP - OFFS	0.6 - 1.4
FALLS	0.4 - 0.6

TABLE 5: CORRECTION BETWEEN FOOT
SENSORIMOTOR FUNCTION AND
BALANCE

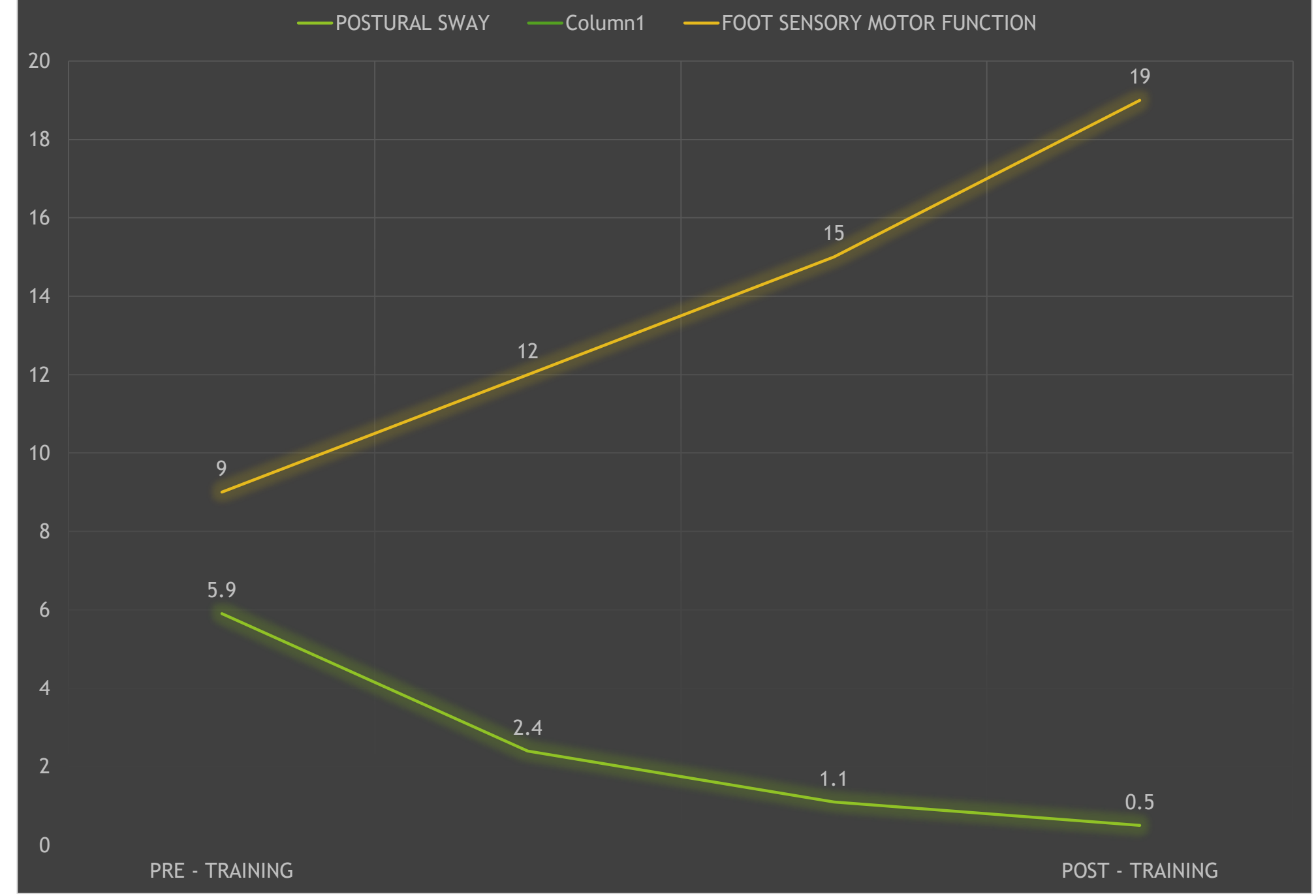
VARIABLE	R- VALUE
Plantar sensitivity vs sway	-0.62
Foot strength vs balance errors	-0.58



DISCUSSION

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- ▶ The results indicate that balance beam performance is strongly associated with postural stability and foot sensorimotor function. Increased postural sway was observed in younger gymnasts, suggesting incomplete neuromuscular control. Gymnasts with higher plantar sensitivity and better foot muscle strength demonstrated fewer balance errors and improved stability.
 - ▶ The negative correlation between plantar sensitivity and postural sway highlights the importance of sensory feedback from the foot in maintaining balance. These findings support previous studies emphasizing the role of somatosensory input in balance regulation. Furthermore, balance errors such as wobbles and step-offs were more frequent in gymnasts with reduced sensorimotor efficiency, reinforcing the need for targeted training interventions.

PRE & POST TRAINING ANALYTICAL GRAPH



CONCLUSION

- This study demonstrates that balance beam performance in young gymnasts requires high postural stability, which is significantly influenced by foot sensory and sensorimotor function. Young gymnasts commonly exhibit balance errors due to immature neuromuscular control and reduced sensory integration. Enhancing foot sensorimotor function through targeted training may improve balance performance and reduce errors during beam activities. Incorporating foot-focused proprioceptive and strengthening exercises into gymnastics training programs may contribute to improved performance and injury prevention.

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THANK YOU